Syllabus

This year's course structure

Lectures

In our lectures, we mostly refrain from showing you any code. Instead, we will focus on explaining the concepts and how they can be used to solve problems. The tutorials will then show you step-by-step how to implement the concepts in Python.

We were rather ambitious in the design of the course in the course module, thus we will not be able to cover all topics there. However, we have chosen the most important topics from our perspective that will lay a good foundation for interested students to continue learning on their own.

Welcome and Optimal Stopping

Lecture I

- Overview of the course and its objectives
- Introduction to algorithmic thinking
- Optimal Stopping
- Programming: First Steps in Python

Explore & Exploit

Lecture II

- Multi-armed Bandit Problem
- Balancing exploration and exploitation (Laplace's rule, Bellman, Gittins index)
- Programming: Lists, loops and simulating slot machines

Caching

Lecture III

- Principles of caching and memory management
- Least Recently Used (LRU) and other caching strategies
- Caching for personal productivity and attention
- Programming: Implementing an LRU cache with lists, functions and packages

Scheduling

Lecture IV

- Task scheduling algorithms (Johnson's rule, EDD, SPT, Lawler)
- Different objectives: deadlines, waiting time, priorities and dependencies
- Real-time scheduling and thrashing

- Programming: Implementing scheduling algorithms with dictionaries and DataFrames

Randomness

Lecture V

- Role of randomness in algorithms
- Monte Carlo methods and their applications in business
- Simulated annealing and the traveling salesman problem
- Programming: Working with AI coding assistants and Monte Carlo simulation

Computational Kindness

Lecture VI

- Computational kindness: algorithms that reduce the burden for others
- Fairness and bias in algorithms
- Quick review of key concepts and outlook (Bayes' rule, overfitting, game theory, networking)
- Brief student presentations
- Open discussion and Q&A